Topology Is Lossy: Skeletonization Discards Discriminative Stroke Information for CNN Digit Recognition

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Abstract

Skeletonization—reducing a stroke to its one-pixel medial axis—is an appealing pre-processing step for handwritten digits: it is stroke-thickness invariant, compact, and classical. We test, in a controlled study, whether it actually helps a convolutional digit classifier, and find a clean negative. Across four standard thinning algorithms (Zhang–Suen, Guo–Hall, Lee, medial-axis), a CNN trained on skeletonized MNIST trails the same CNN on raw pixels by 0.9–1.2 percentage points (best skeleton 98.11% vs raw 99.03%; 3 seeds). The loss is not recoverable: adding the skeleton as a second input channel alongside the raw image (*fusion*) does *not* beat raw pixels alone (98.98% vs 99.03%), and adding Hough line features to the skeleton changes accuracy by $+0.01$ pp. The degradation concentrates on stroke-width- and density-sensitive digits (3, 5, 9, 8), and the most expensive skeletonizer (medial-axis, $100\times$ slower) is no more accurate. The takeaway is simple and, we argue, useful to state plainly: for digit recognition, stroke width and grayscale intensity carry discriminative information that thinning throws away, and a topological representation alone is strictly worse—a result that should temper the recurring intuition that skeletons are a good front-end for digit CNNs.

1 Introduction

Thinning a binary shape to its medial axis [1] is a classical operation with an intuitive appeal for handwriting: it normalises stroke thickness, yields a compact graph-like representation, and has fast parallel algorithms [7, 3]. It is therefore a recurring idea to feed *skeletonized* digits to a neural classifier. We ask the direct empirical question—does it help?—and report a controlled negative: on MNIST [4], skeletonization consistently *hurts* a CNN, the loss is not recovered by fusing the skeleton with the raw image, and it traces to information (stroke width, intensity) that thinning is designed to remove. Negative results like this are worth recording: the intuition that "topology is enough" for digits is common, and a clean controlled refutation saves re-implementation effort.

2 Setup

We use one backbone (a small two-block CNN, `SimpleCNN`) and vary only the input. **Inputs:** raw 28×28 grayscale; skeletons from four algorithms—Zhang–Suen [7], Guo–Hall [3], Lee [5], and a medial-axis transform—each binarised at a fixed threshold; optional Hough-line channels [2]; and a two-channel *fusion* of raw + Guo–Hall skeleton. All configurations share epochs, batch size, learning rate, and architecture; we report mean $\pm$ std test accuracy over 3 seeds.

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Table 1: MNIST test accuracy (%, mean $\pm$ std over 3 seeds) and skeletonization cost. Every topological input trails raw pixels; fusion does not recover the gap.

Input	Channel	Accuracy	Skel. time (s)
Raw pixels	raw	$\mathbf{99.03 \pm 0.06}$	—
Guo–Hall [3]	skeleton	98.11 ± 0.09	26
+ Hough [2]	skeleton+hough	98.12 ± 0.09	26
Lee [5]	skeleton	98.02 ± 0.07	8
medial-axis	skeleton	97.95 ± 0.11	2978
Zhang–Suen [7]	skeleton	97.81 ± 0.04	—
Raw + Guo–Hall (fusion)	raw+skeleton	98.98 ± 0.05	26

Table 2: Per-class accuracy (%, mean over 3 seeds): raw vs. Guo–Hall skeleton vs. fusion.

	0	1	2	3	4	5	6	7	8	9
raw	99.7	99.5	98.9	99.3	99.1	99.0	98.7	99.3	98.8	97.9
skeleton	99.8	99.2	98.5	97.6	97.9	97.4	98.0	98.2	97.6	96.7
fusion	99.7	99.8	99.0	99.2	99.3	98.9	98.3	99.2	98.3	98.1

3 Results

Every skeletonizer underperforms raw pixels (Table 1, Fig. 1). The best skeleton (Guo–Hall, 98.11%) trails raw pixels (99.03%) by 0.92 pp; the worst (Zhang–Suen, 97.81%) by 1.22 pp. The gap exceeds the seed-to-seed std (≤ 0.11) by an order of magnitude, so it is not noise.

The loss is not recoverable. Two controls rule out "the skeleton adds complementary information the CNN just needs help using." (i) **Fusion**: giving the network *both* the raw image and its skeleton as two channels yields 98.98%—statistically indistinguishable from, and numerically below, raw alone (99.03%). The skeleton channel contributes nothing the pixels do not already carry. (ii) **Hough lines**: appending Hough line-segment features to the skeleton moves accuracy by $+0.01$ pp across all four thinners—within noise.

Where the information goes (Table 2, Fig. 1 right). The degradation is not uniform: digits 0 and 1 are essentially unaffected, while 3 (-1.7 pp), 5 (-1.6), 9 (-1.2), 4 (-1.2) and 8 lose the most under Guo–Hall; under Zhang–Suen, digit 8 collapses by 3.1 pp ($98.8 \to 95.7$). These are exactly the digits whose identity depends on stroke *thickness*, loop *density*, and intersection detail (open vs. closed loops, doubled strokes)—precisely what one-pixel thinning erases. Fusion's per-class accuracy tracks raw, confirming the skeleton channel adds nothing.

Cost. The medial-axis transform is $\sim 100\times$ slower than Guo–Hall (2978 s vs. 26 s to skeletonize the dataset) and *less* accurate—there is no accuracy/cost frontier on which any skeletonizer dominates raw pixels.

4 Discussion

The result is unsurprising in hindsight but worth stating because the opposite intuition is common. Skeletonization is *designed* to be invariant to stroke width; for digit *recognition*, stroke width and grayscale intensity are not nuisances but *features* (they separate a thick-looped 8 from a 0, a heavy 5 from a 6, etc.). Discarding them is a strict information loss that a CNN—which already learns its own thickness-tolerant features from raw pixels [6]—cannot recover, and that re-supplying the skeleton as a side channel does not help. We emphasise the *controlled* nature of the claim: same backbone, same training, four thinners, a fusion control, and a Hough control all point the same way.

Scope. This is a statement about *supervised, data-rich* digit recognition with a CNN. It does *not* say skeletons are useless in general: thickness-invariant topological descriptors can be competitive in

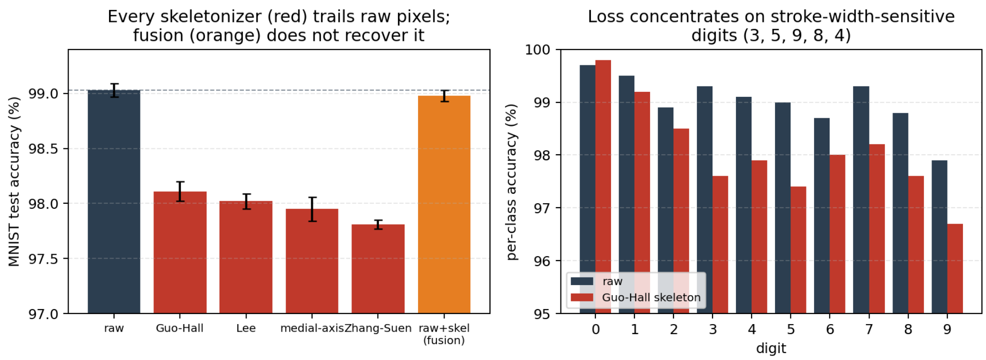

Figure 1: Left: every thinning algorithm (red) trails raw pixels (dashed line); the raw+skeleton fusion (orange) does not recover the gap. Right: per-class accuracy, raw vs. Guo–Hall skeleton; the loss concentrates on stroke-width/density-sensitive digits.

the extreme low-data / one-shot regime and can add robustness under stroke-width corruption, where the CNN's pixel features are unreliable. The negative is specifically that, when you have pixels and labels, thinning them first only throws information away.

5 Conclusion

For supervised CNN digit recognition, skeletonization is a lossy front-end: across four thinning algorithms it costs $\sim$1 pp versus raw pixels, the loss concentrates on stroke-width-sensitive digits, and it is recovered neither by raw+skeleton fusion nor by Hough line features. Topology alone is not enough; keep the pixels.

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Reproducibility: `scripts/run_skeleton_experiment.py` (all configurations, 3 seeds); raw numbers in `experiments/reports/skeleton_comparison.md`; skeletonizers in `src/skeleton/`.

References

[1] Harry Blum. A transformation for extracting new descriptors of shape. *Models for the Perception of Speech and Visual Form*, pages 362–380, 1967.

[2] Richard O. Duda and Peter E. Hart. Use of the Hough transformation to detect lines and curves in pictures. *Communications of the ACM*, 15(1):11–15, 1972.

[3] Zicheng Guo and Richard W. Hall. Parallel thinning with two-subiteration algorithms. *Communications of the ACM*, 32(3):359–373, 1989.

[4] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[5] Ta-Chih Lee, Rangasami L. Kashyap, and Chong-Nam Chu. Building skeleton models via 3-D medial surface axis thinning algorithms. *CVGIP: Graphical Models and Image Processing*, 56 (6):462–478, 1994.

[6] Patrice Y. Simard, Dave Steinkraus, and John C. Platt. Best practices for convolutional neural networks applied to visual document analysis. In *International Conference on Document Analysis and Recognition (ICDAR)*, 2003.

[7] T. Y. Zhang and Ching Y. Suen. A fast parallel algorithm for thinning digital patterns. *Communications of the ACM*, 27(3):236–239, 1984.